Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110
(An Autonomous Institution, Affiliated to Anna University, Chennai)
Department of Computer Science and Engineering
Model Practical Examination

Degree & Branch	M.Tech (Integrated) Computer Science & Engineering Semester V
Subject Code & Name	ICS1512 – Machine Learning Algorithms Laboratory
Academic Year	2026–2027 (Odd) Batch: 2024–2029
Date	

Evaluation Scheme

	a	b	c	d	e	f
	EDA	**Regression Model**	**PCA**	**Evaluation**	**Statistical Analysis**	**Best Practices & Learning Outcomes**
Max Marks	15	15	20	20	10	20
Marks Scored						

Dataset

Use the supplied **Machine Remaining Useful Life Regression** dataset.

Dataset link: `REGRESSION_DATASET_DRIVE_LINK`

The dataset contains **1,000 observations**. The continuous target variable is `Remaining_Useful_Life_Hours`. Predictors include numerical machine/sensor measurements, categorical variables (`Machine_Type` and `Shift`), a unique `Record_ID`, missing values, mild outliers, and correlated features.

For the assigned regression algorithm, students must implement:

- **Model–1:** regressor **without PCA**;
- **Model–2:** the **same regressor with PCA**.

Use the **same 5-Fold Cross-Validation splits** for both models so that the fold-wise validation errors are directly paired.

Develop a regression model to predict `Remaining_Useful_Life_Hours` using **Support Vector Regression (SVR) with RBF kernel**. The objective is to **compare** the same regressor **with and without PCA** using identical cross-validation folds and to determine whether the observed difference is statistically significant.

Tasks

(a) **Analyze** the dataset characteristics by examining data types, missing values, descriptive statistics, skewness, target distribution, predictor distributions, correlations, and outliers. **Identify** the non-predictive identifier and **apply** appropriate preprocessing including imputation, categorical encoding, and justified outlier treatment. **[15 Marks]**
CO1; K3–K4; PIs: 1.3.1, 1.4.1, 3.4.2, 4.3.3

(b) **Apply Support Vector Regression (SVR) with RBF kernel** to the preprocessed feature representation **without PCA**. Use a single RBF-kernel SVR formulation for this experiment. **Select**

suitable hyperparameters using a leakage-free tuning strategy. Tune C, γ, and ϵ using an appropriate grid or randomized search. Report the final parameter configuration and baseline regression model. **[15 Marks]**
CO1; K3; PIs: 2.3.1, 3.4.2, 14.7.1

(c) **Analyze** correlation and redundancy among numerical predictors and **apply** Principal Component Analysis after appropriate scaling. Determine the minimum number of components required to retain at least **95% cumulative explained variance**. Present the explained variance and the original versus reduced dimensions. Then **apply the same Support Vector Regression (SVR) with RBF kernel** to the PCA-transformed data using the same modelling protocol used in part (b). **[20 Marks]**
CO2; K3–K4; PIs: 4.2.1, 4.3.3, 5.1.2

(d) **Evaluate** Model–1 and Model–2 using the **same 5-Fold Cross-Validation splits**. Record the validation RMSE for each fold:

Fold	RMSE Without PCA	RMSE With PCA	RMSE Difference
1			
2			
3			
4			
5			

For both models, report mean and standard deviation of RMSE together with MAE, R^2, and MAPE. Also examine whether PCA changes the gap between training and validation errors. **Compare** the two representations and **justify** which gives better predictive generalization. **[20 Marks]**
CO2, CO3; K4–K5; PIs: 2.2.4, 4.3.3, 14.7.2

(e) Using the **five paired fold-wise RMSE values** from part (d), **evaluate** statistical significance between the with-PCA and without-PCA models using a **two-sided paired t-test**. Define

$$H_0 : \mu_d = 0 \qquad \text{and} \qquad H_1 : \mu_d \neq 0,$$

where d is the paired fold-wise RMSE difference. Compute the t-statistic and p-value and **interpret** the result at $\alpha = 0.05$. State whether the null hypothesis is rejected or not rejected. **[10 Marks]**
CO3; K5; PIs: 4.2.1, 14.7.2

(f) **Design** a final leakage-free regression workflow and **propose** whether the **with-PCA** or **without-PCA** representation should be adopted. Prepare a structured comparison covering preprocessing, dimensionality, cross-validation consistency, RMSE/MAE/R^2, statistical-test conclusion, overfitting/generalization, computational cost, interpretability, and reproducibility. **Evaluate** the limitations of the final solution and **propose** at least two meaningful improvements. Summarize the major learning outcomes obtained from the experiment. **[20 Marks]**
CO1, CO2, CO3; K5–K6; PIs: 3.1.1, 3.2.1, 10.3.1, 14.7.1

Important: Imputation, encoding, scaling, PCA, and hyperparameter selection must be performed **inside the training folds**. The **same regressor and same five folds** must be used for the PCA and non-PCA models.